



BUSINESS ANALYSIS REPORT

client
Popular Auto Spare and A/C Parts Co. W.L.L



Consultant

UNIGLOBAL
CONSULTANCY

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PREFACE

This document presents the Business Analysis Study for the PCode Generation Module. It has been prepared based on a series of discussions conducted with the team of **Popular Auto Spare & A/C Parts Co. W.L.L** to understand the existing process, workflow, and requirements related to **PCode** generation through purchase activities and new request from the branches.

The purpose of this document is to capture the functional requirements, process flow, and key parameters identified during these discussions. It reflects the current understanding of the module's behavior, roles, responsibilities, and system interactions as shared by the concerned teams.

This document is intended to serve as the first version of the functional understanding of the PCode Generation process. It does not include the development timeline, effort estimation, or cost for the software implementation of this module. These aspects will be finalized only after the review and confirmation of this document, along with the detailed functionality, by the consultant and the management of Popular Automobiles.

Once the workflow and functional requirements are formally approved, a separate document will be provided outlining the development timeline, cost estimation, and implementation plan for the module.





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1. ABOUT SANDS LAB

SaNDS Lab (Software and Network Development Solutions Lab) is a leading IT solutions provider with **16 years** of expertise in custom software development, IT training, and digital transformation services. Headquartered in India, Trichur Infopark, Kerala, Koratty with a strong presence in Bahrain as **SaNDS Lab Middle East W.L.L.**, we specialize in delivering innovative, scalable, and reliable technology solutions across various industries. For the past 5 to 6 years, we have been serving our clients in Bahrain as a trusted outsourcing partner. Now we have a team of localized support personnel, forging stronger relationships with our clients in the region.

At **SaNDS Lab**, we specialize in creating tailored software solutions that meet the unique needs of our clients. Our team of skilled professionals is dedicated to ensuring that every project we undertake is executed with the highest standards of excellence and precision. Our broad range of services and innovative approach have earned us a reputation for reliability and excellence in the software development industry.

Our core competencies include:

Custom Software Development – Tailor-made solutions to meet specific business needs.

1. Web & Mobile App Development – Creating dynamic, user-friendly applications.
2. Blockchain & FinTech Solutions – Secure and decentralized applications for modern businesses.
3. Headless CMS & API Development – Scalable and efficient content management systems.
4. IT Training & Certification – Bridging the gap between industry and fresh talent.
5. Cloud & DevOps Services – Streamlined deployment and infrastructure management.

2. Our Mission

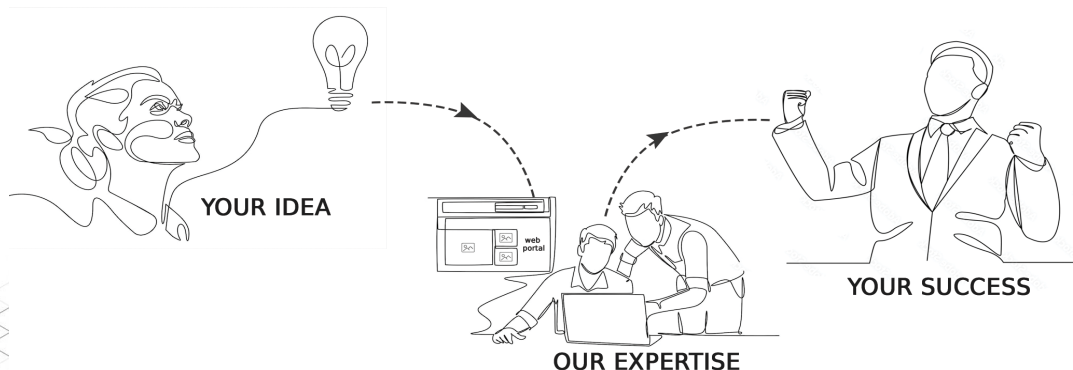
At SaNDS Lab, we are committed to providing cutting-edge technology solutions that empower businesses to grow and succeed.

Our motto,

“Your Idea, Our Expertise, Your Success”,

reflects our dedication to delivering excellence and innovation.

“ Delivering the Right Solutions at
the Right Time to the Right People ”



3. Business Background

Popular Auto Spare and A/C Parts Co. is a multi-branch trading company headquartered in Bahrain, with operational branches across Bahrain and Dubai. The company has a strategic expansion plan to grow across global markets. All operational, financial, and administrative governance is centralized at the Bahrain head office to ensure consistency, compliance, and effective control across all branches.

The organization operates in core areas including sales, procurement, inventory management, stock movement, human resources, finance, and accounting. With approximately 50-80 employees worldwide, the company currently depends on **BUSY ERP** software supplemented by multiple third-party tools to manage daily operations. While these systems provide functional support, they create fragmented workflows and require heavy manual intervention.

4. Current Challenges / Problem Statement

The current operational environment relies on a combination of legacy ERP software and multiple third-party tools, resulting in process fragmentation and operational inefficiencies. While the system supports basic day-to-day activities, it does not provide the level of integration, automation, and governance required for a growing multi-branch organization.

4.1. Key challenges are outlined below:

- **High Dependency on Individual Employee Expertise**

Many operational tasks rely heavily on the personal knowledge and experience of specific employees rather than standardized system-driven workflows. Critical processes are often managed through informal practices, spreadsheets, or undocumented procedures. This creates operational risk when employees are absent or leave the organization, as knowledge transfer is limited and replacement staff require significant training time. The business becomes person-dependent instead of process-dependent.

- **Manual Processes Increasing Workload**

Several operational activities involve manual data entry, reconciliation, and cross-checking between systems. These repetitive tasks increase employee workload, consume valuable time, and raise the likelihood of human error. Instead of focusing on strategic or customer-oriented tasks, employees spend significant effort maintaining operational records.

4.2. Difficulty Maintaining Operational Continuity During Employee Turnover

When experienced staff leave or change roles, operations may slow down or temporarily disrupt because procedures are not fully system-driven. Replacement employees must learn complex manual processes, increasing onboarding time and operational risk. The company's ability to function smoothly should not depend on specific individuals.

4.3. Reduced Visibility of Stock Availability and Shortage Monitoring

Management at headquarters currently lacks real-time visibility into stock availability across branches. Inventory data is not fully synchronized in a centralized system, making it difficult to accurately monitor stock levels, identify low-stock items, or track fast-moving and slow-moving inventory.

Stock status reporting often requires manual consolidation from multiple sources, which delays decision-making and increases the risk of stockouts or overstocking. Limited visibility affects procurement planning, branch replenishment, and inter-branch stock transfers. As a result, the organization may face situations where critical items are unavailable in one branch while excess stock exists in another.

The absence of centralized, real-time inventory insight reduces the ability of management to optimize stock levels, forecast demand accurately, and maintain consistent product availability across all locations.

4.4. Scalability Constraints for Global Expansion

The current system architecture is not optimized for rapid multi-branch scaling. As the company expands into new regions, managing additional branches using the existing fragmented tools will increase complexity, operational risk, and administrative overhead. A scalable platform is essential to support planned international growth.

4.5. Overall Impact

These challenges collectively reduce productivity, increase operational risk, and limit the organization's ability to grow efficiently. Without a centralized, integrated ERP solution, the company faces increasing complexity as it expands, which may result in higher costs, inconsistent operations, and reduced competitive advantage.

CURRENT CHALLENGES

MANUAL WORK



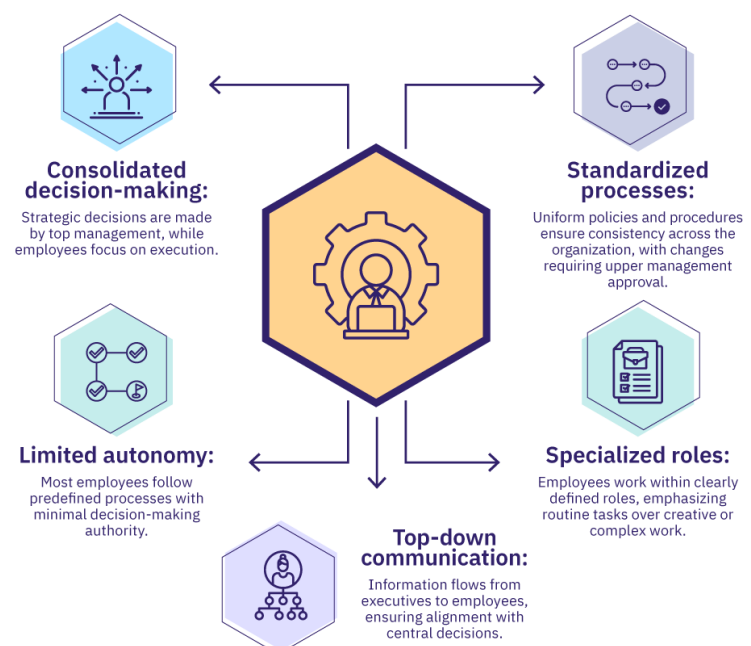
PROCESS DELAY



INCORRECT INFORMATION



Key Features of Centralized Management



5. Business Objectives

The implementation of a new ERP system is a strategic initiative aimed at transforming Popular Auto Spare and A/C Parts Co. into a process-driven, scalable, and globally competitive organization. The objectives extend beyond software replacement and focus on long-term operational maturity, governance, and growth enablement.

5.1. Establish a Centralized, Fully Integrated Operational Platform

The organization seeks to unify all departments — sales, procurement, inventory, HR, finance, and administration — within a single integrated system. A centralized ERP will serve as the single source of truth, ensuring consistent data, eliminating duplication, and improving cross-department coordination. This integration will allow management to monitor operations across all branches in real time and enforce standardized policies from headquarters.

5.2. Reduce Employee Workload Through Automation

Automation of inventory monitoring and procurement processes is a core objective of the proposed ERP. The system should automatically track low-stock items, support efficient stock verification in stores, and generate alerts for replenishment requirements. By systemizing the preparation of purchase requests, supplier quotations, and purchase orders, the ERP will eliminate repetitive manual coordination and paperwork.

Employees will no longer need to spend excessive time identifying shortages, verifying stock manually, or preparing procurement documents from scratch. Instead, they can focus on supplier negotiation, inventory optimization, and customer service. Automation will improve accuracy in stock records, reduce delays in purchasing cycles, and ensure timely availability of critical items across branches while minimizing human error and operational workload.

5.3. Improve Productivity and Operational Efficiency

The ERP is expected to streamline workflows and reduce process bottlenecks. Faster transaction processing, automated reconciliation, and integrated reporting will enhance departmental productivity. Efficiency gains will allow the company to handle increased business volume without proportional increases in workforce.

5.4. Standardize Processes Across All Branches

A unified ERP platform will enforce consistent workflows, approval structures, and reporting formats across all locations. Standardization ensures that every branch operates under the same operational framework, improving accountability, audit readiness, and management control.

5.5. Strengthen Organizational Hierarchy and Administrative Structure

The new system will support defined roles, responsibilities, and approval hierarchies. This structure ensures transparency in decision-making, proper authorization controls, and clear accountability. A well-defined hierarchy reduces ambiguity and improves governance.

5.6. Enable Seamless Operational Continuity Independent of Employee Changes

A system-driven environment ensures that knowledge is embedded in processes rather than individuals. When employees transition roles or leave the company, operations continue without disruption because procedures are documented and enforced by the ERP. This reduces operational risk and dependency on specific personnel.

5.7. Align Employee Roles with Their Experience and Potential

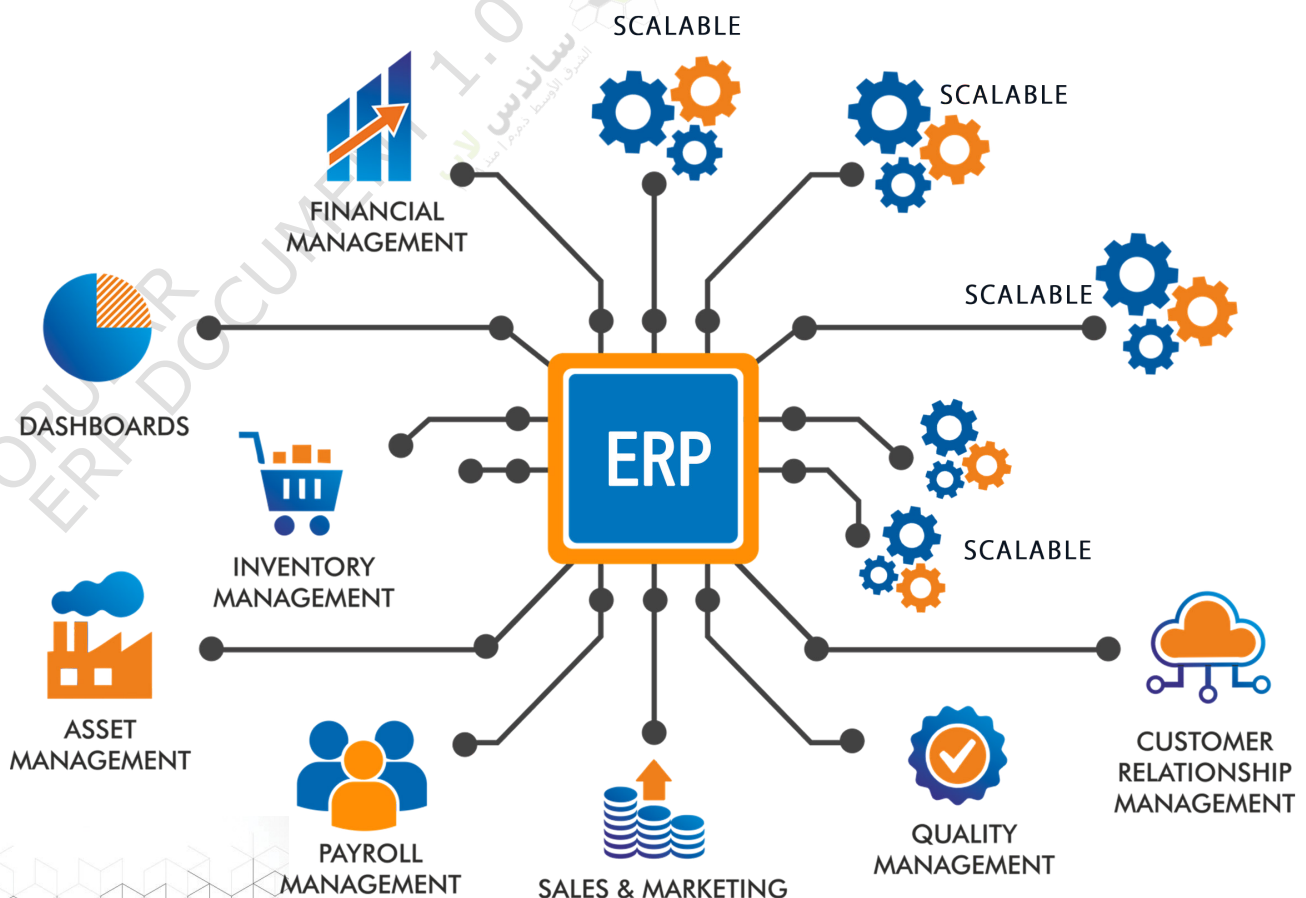
By reducing routine administrative burden, employees can focus on tasks aligned with their skills and expertise. The ERP will enable better role allocation, performance tracking, and workforce optimization, helping management utilize human resources more strategically.

5.8. Support Business Expansion and Diversification

The company's growth strategy requires a system capable of managing additional branches, currencies, tax structures, and regulatory requirements. The ERP will provide a scalable operational backbone that supports international expansion without compromising control or efficiency.

5.9. Create a Scalable System Suitable for Global Growth

The proposed ERP must be designed for long-term scalability. As transaction volumes, branch networks, and workforce size increase, the system should continue to perform reliably. Scalability ensures that technology will not become a limiting factor in business growth.



6. Vision for the Proposed ERP

Management envisions the proposed ERP system as a foundational platform that will redefine how the organization operates, collaborates, and grows. The ERP is intended to function as a single, reliable source of truth across all departments, ensuring that every operational, financial, and administrative decision is based on accurate, real-time data.

The system is not viewed merely as a tool for automation, but as a strategic enabler that elevates the company to a more structured, process-oriented, and governance-driven organization. By embedding standardized workflows, approval hierarchies, and compliance controls into daily operations, the ERP will promote discipline, transparency, and accountability at every level of the company.

A key element of the vision is to transition from person-dependent operations to system-dependent processes. Business continuity should be guaranteed by the system itself, ensuring that operations continue smoothly regardless of staff changes, branch expansion, or increases in transaction volume. This shift will reduce operational risk and strengthen institutional knowledge within the organization.

The ERP is also expected to support management with real-time analytics, performance monitoring, and consolidated reporting across all branches. Enhanced visibility will enable faster, data-driven decision-making and improve strategic planning. With centralized oversight and standardized data, management can proactively respond to market changes and operational challenges.

In the long term, the ERP will serve as the technological backbone for global expansion. The system should support multi-branch, multi-currency, and multi-regulatory environments, allowing the company to scale confidently into new markets. The vision is to create a resilient digital infrastructure that supports sustained growth, operational excellence, and competitive advantage.

Ultimately, the proposed ERP aims to transform the organization into a mature, process-driven enterprise where efficiency, accountability, and scalability are built into the operational framework, ensuring long-term stability and sustainable growth.

7. Multi-Company and Branch Structure with ERP Operational Workflow

This section defines the organizational structure of Popular Automobiles and explains how the proposed ERP system will support a multi-company, multi-branch operating model. The framework ensures centralized governance from headquarters while enabling scalable branch operations across countries.

7.1. Company and Division Overview

Popular Auto Spare and A/C Parts Co. commenced operations in 2003, focusing on automotive spare parts and A/C components. In the same year, the company expanded its service offerings by establishing Smart Popular Motor Services — a professional automotive garage in Salmabad providing comprehensive mechanical and A/C services for passenger vehicles and commercial fleets.

To cater to industrial clients, Popular Industrial Parts was launched in 2009 at the Sitra branch.

7.2. Branch Network

1. Automobile Spare Parts Division: 6 branches — 5 in Bahrain and 1 in Dubai. The Dubai branch currently operates wholesale only, with retail operations on hold.
2. Industrial Parts Division: 2 branches in Bahrain
3. Motor Services Division: 1 branch in Salmabad

The ERP implementation will initially focus on the Automobile Spare Parts division, standardizing operations across all six branches. With this foundation, the company will be able to expand into additional countries while maintaining consistent processes and centralized control. After successful rollout, the ERP will scale to include the Industrial Parts and Motor Services divisions, forming a fully integrated multi-company platform.

7.3. ERP Company Creation Framework

The ERP system will support the creation of company entities with defined organizational attributes, including:

1. Company name
2. Country of operation
3. Base currency & exchange rate with Bahraini Dinars
4. VAT / tax configuration
5. Regulatory and financial settings

These attributes serve as default standards inherited by all branches under the company.

7.4. Branch Creation and Structure

Each company can contain multiple branches configured with:

1. Branch name
2. CR Number
3. Physical address
4. Auto-generated branch code
5. Country assignment
6. Classification as Store or Outlet

Branches automatically inherit company-level financial and regulatory rules, ensuring operational consistency. Within each country, the ERP supports central stores and multiple outlets to enable structured inventory distribution.

7.5. Centralized Procurement Workflow

Procurement activities across countries and branches will be centrally managed by the Bahrain headquarters:

1. Branches generate purchase requests

2. Requests are reviewed centrally
3. Supplier quotations are managed from Bahrain
4. Purchase orders are issued from Bahrain
5. Goods are allocated to relevant branches

This model ensures pricing control, supplier governance, and standardized procurement.

7.6. User Hierarchy and Access Control

The ERP supports structured user roles across branches and countries:

- **Role-based permissions**
- **Branch and country-level access control**
- **Approval hierarchies**
- **Segregation of duties**

This framework ensures accountability, security, and compliance.

7.7. Scalability and Expansion Support

The ERP architecture enables rapid onboarding of new branches or countries without system restructuring. New entities inherit standardized company settings, ensuring operational continuity and future scalability.

8. Inventory Booking and PCode (Popular Code) Management Module

This module defines the inventory identification structure, PCode lifecycle, substitute mapping framework, and governance controls required to manage automotive spare parts inventory across multi-branch operations. The objective is to create a controlled, scalable, and system-driven item management environment that eliminates external dependencies and preserves organizational trade intelligence.

8.1. Current Inventory Structure

Popular Auto Spare and A/C Parts Co. manage inventory using an internally defined identifier known as PCode (Popular Code).

A PCode is a unique alphanumeric identifier assigned to automotive parts or items. It is designed to capture minute differences between products, such as:

- **OEM (Original Equipment Manufacturer) variations**
- **Part numbers**
- **Brand differences**
- **Other relevant parameters**

While the automotive industry standard relies on OEM numbers and manufacturer part numbers, the organization uses PCode which can accurately track and manage parts, even when they are very similar in function or specification. This ensures precision in inventory management, sales, procurement, and



compatibility mapping in the automotive industry.

In the organization, PCodes are generated sequentially using a FIFO (First In, First Out) method, ensuring that each new item receives the next available code in the sequence.

Each item is categorized under a defined Group, enabling structured classification and reporting.

Group Master Parameters:

- **Group name**
- **Status (Active / Inactive)**
- **The Item Master currently maintains detailed attributes including:**
 - **PCode**
 - **Item name**
 - **Group name**
 - **Alias**
 - **OEM number (supports multiple references)**
 - **Manufacturer part number**
 - **Brand**
 - **Country of origin**
 - **Vehicle model**
 - **Engine number**
 - **Engine CC**
 - **Position / fitment**
 - **Size / specification**
 - **Remarks**
 - **Purchase unit**
 - **Sales unit**
 - **Tax percentage**
 - **Purchase price**
 - **Sales price**
 - **MRP**
 - **Minimum sales price**

All pricing is tax-inclusive. Pricing structures are country-specific and maintained in local currency. The system supports automatic currency conversion with controlled user override. PCode visibility across countries is governed at the master configuration level.

The system also supports attachment of multiple product images within the item master, enabling visual reference during sales and procurement.

8.2. Current Operational Challenges

The existing PCode creation process is partially external to the ERP and relies on Excel-based workflows. This creates operational risk and inefficiency:

- **Dependence on third-party tools**
- **Manual bulk upload procedures**
- **Risk of duplicate or inconsistent codes**
- **Difficulty in price updtaiions**
- **Sequence integrity issues**
- **Data loss risk during entry**
- **Legacy unused PCodes affecting search performance**
- **Lack of controlled approval workflow**

These limitations increase operational dependency on individuals and reduce data governance.

8.3. Proposed Enhancement – Integrated PCode Generation Module

The proposed ERP will introduce a fully integrated PCode Generation Module, eliminating external spreadsheet dependency and ensuring controlled item onboarding.

Core Capabilities:

- **Automated system-generated PCode sequencing**
- **System suggestion of next available code**
- **Optional user override with validation**
- **Real-time uniqueness verification**
- **Item-wise save functionality to prevent data loss**
- **Visual progress indicators:**
 - **Unsaved** ☐ **Red**
 - **Saved** ☐ **Green**
- **Immediate master visibility after submission**
- **Editable item attributes post-creation**
- **Permanent lock on PCode after generation**

Once assigned, a PCode becomes a permanent identifier and cannot be modified. This ensures long-term traceability and audit integrity.

8.4. Business Purpose of PCode

The PCode (Popular Code) serves as the primary internal identifier for all items within Popular Auto Spare and A/C Parts Co.'s ERP system. Beyond simple identification, the PCode framework provides several critical businesses, operational, and strategic benefits:

- **Unique Internal Identification:**

Each item in inventory is assigned a distinct PCode, ensuring no duplication or ambiguity across branches or countries.

Facilitates accurate tracking of stock movement, purchase orders, sales transactions, and returns.

Supports inventory reconciliation and audit processes by providing a stable reference independent of OEM or manufacturer numbering systems.

• Pricing Strategy and Confidentiality Protection

The PCode enables internal control over pricing by decoupling external OEM numbers from sales and procurement documents.

Prevents competitors or unauthorized personnel from deducing the company's pricing strategy.

Allows flexible country-level pricing while maintaining centralized governance.

• Preservation of Trade Secrecy

PCodes act as a protective mechanism for proprietary operational knowledge, especially in sales and part substitutions.

Critical for safeguarding internal processes, substitute mapping logic, and technical specifications.

• Standardization of Operational Workflows

PCodes provide a consistent reference across sales, procurement, inventory, finance, and reporting modules.

Ensures uniform workflows in multi-branch operations, reducing errors and miscommunication.

Enables automation in approvals, stock transfers, and reporting without dependency on staff memory or experience.

• Prevention of OEM Exposure in Commercial Documents

All sales documents display PCode instead of OEM numbers.

Prevents sensitive supplier and pricing information from being exposed externally.

Supports compliance with internal confidentiality and competitive intelligence policies.

• Consistent Reporting and Analytics

PCodes provide a stable reference for data aggregation and analytics, enabling accurate sales trends, inventory performance, and procurement insights.

Supports multi-dimensional reporting, including brand, vehicle model, branch, and price segment analysis.

Facilitates strategic decision-making for stock replenishment, promotions, and expansion planning.

Objective: The PCode framework is not merely a technical identifier—it is a strategic operational tool. By providing unique identification, protecting pricing and trade secrecy, standardizing processes, and enabling accurate reporting, PCodes form the backbone of a secure, standardized, and strategically governed operational framework.

8.5. PCode Request Approval Workflow

A formal multi-level approval workflow ensures that no item enters inventory without validation:

- 1. Request Submission:** Sales Manager/ Store Supervisor created PCode request in ERP, with all item details captured. Status: Draft / Pending.
- 2. CTO Technical Validation:** Reviews OEM compatibility, specifications, pricing strategy, substitute mapping, and compliance. May edit attributes before approval.
- 3. Final Approval:** CTO approves or rejects. Upon approval, ERP generates official PCode, which becomes permanent and non-editable.
- 4. Inventory Activation:** Item becomes active in inventory. Transactions referencing the item are now permitted.

Governance Rule: No purchase, stock movement, or sales transaction is allowed without an approved PCode.

8.6. Exception Handling for Items Without PCode Registration

To manage urgent operational scenarios while maintaining governance and data integrity, the proposed ERP will introduce a dedicated non-PCode Item Master. This master acts as a controlled holding area for items that are temporarily handled in operations before formal PCode registration.

Non-PCode Item Master Workflow

1. Temporary Item Registration

- Items handled during emergency sales or urgent store requirements are recorded in a dedicated non-PCode Item Master.
- These items remain operationally visible but are clearly flagged as unregistered.
- Transactions involving such items are traceable and linked to the temporary record.

2. CTO Review Dashboard

- The CTO is provided with a structured dashboard listing all pending non-PCode items.
- Each entry includes item details, transaction references, and submitted specifications.
- This ensures centralized visibility and prioritization of pending registrations.

3. PCode Generation by CTO

- The CTO selects an item from the pending list and completes the required item attributes and mapping information.
- Upon confirmation, the ERP generates a PCode in strict sequential order based on system rules.
- The item is automatically migrated from the non-PCode master to the official Item Master.

4. Item Deletion Control

- If an item is determined to be invalid, duplicated, or unnecessary, the CTO has authority to remove it from the non-PCode master.

All deletions are audit-logged for governance and traceability.

5. Transaction Synchronization

Once a PCode is generated, all temporary transactions are automatically linked to the official item record. Inventory, sales, and reporting modules update accordingly.

8.7. Legacy PCode Governance

The ERP will include a structured mechanism to manage historical or inactive PCodes, ensuring that the master data remains clean, searchable, and operationally efficient while maintaining audit compliance.

Key Features and Procedures:

1. Hiding Inactive PCodes

PCodes that are no longer in use (e.g., discontinued parts, obsolete stock, or duplicates) can be marked as Inactive. Inactive PCodes are automatically hidden from all operational searches and transaction screens. Users can only view inactive PCodes through a dedicated administrative or audit interface.

2. Search Optimization

Hidden PCodes are excluded from search results in sales, inventory, and purchase modules. Reduces clutter and ensures employees find only active, valid items. Improves system response time and search performance across large datasets.

3. Master Data Integrity

Cleaning inactive PCodes preserves the accuracy of the Item Master. Reduces risk of errors during sales, procurement, and inventory reconciliation. Prevents accidental selection of obsolete items in transactions.

4. Audit and Compliance Retention

All legacy PCodes remain archived within the ERP for compliance and audit purposes. Historical data is fully traceable, including original creation date, usage history, and deactivation reason. Supports internal audits, statutory reporting, and retrospective analysis.

5. Operational Benefits:

- Streamlines daily operations and reduces user confusion.
- Enhances ERP performance and reporting efficiency.
- Enables management to focus on current, active inventory items.
- Supports continuous improvement in master data governance.

Objective: This structured cleanup ensures that the ERP maintains a clean, efficient, and auditable inventory database, while still preserving historical information for compliance, reporting, and analytical purposes.

8.8. Price Updation Through PCode

Current Scenario

In the current system, updating the price of a PCode is a time-consuming and repetitive process. The responsible personnel must first update pricing details in Excel and then upload the file separately into each country database, resulting in inefficiency and duplication of effort.

- Significant operational effort
- Risk of inconsistency between countries
- Delays in price synchronization
- Increased dependency on manual intervention
- Higher chance of human error

Although the system allows item search across master parameters, the update process remains fragmented and inefficient.

Proposed ERP Enhancement – Centralized Price Management

The proposed ERP introduces a centralized price management interface linked directly to the PCode master.

Key Functionality:

- Advanced search capability across all item parameters
- Unified listing of all eligible items matching the search
- Single-screen price editing interface
- Visibility of country-wise pricing in one view
- Editable pricing per country when required
- Bulk update confirmation through a single action

When the user updates and confirms pricing:

- The system automatically synchronizes prices across all linked country databases
- Currency conversion rules are applied where necessary
- Country-specific overrides are preserved if configured

Control and Governance

- Only authorized roles can modify pricing
- All price updates are audit-tracked
- Historical price versions are stored
- Approval workflow can be enabled if required
- Change logs available for management review





Business Benefits

- Eliminates repetitive manual updates
- Ensures pricing consistency across countries
- Reduces operational workload
- Improves speed of pricing decisions
- Strengthens governance and audit control
- Supports centralized commercial strategy

Summary:

Centralized PCode price management transforms pricing from a manual, fragmented activity into a controlled, efficient, and synchronized process, enabling faster decisions while maintaining accuracy and compliance.

8.9. Minimum Order Level and Centralized Stock Review Control

The proposed ERP will support a Minimum Order Level configuration for each item at both store and branch level, enabling proactive stock monitoring and preventing stock shortages. This feature ensures that critical and fast-moving items are replenished in a timely and controlled manner.

Minimum Level Configuration

Authorized personnel (Sales Manager / Inventory Controller) can define a minimum stock threshold per item for each branch

- Separate thresholds can be maintained for central warehouses and branch stores
- Levels can be adjusted based on consumption trends and seasonal demand
- Automated Alert Mechanism
- When stock falls below the configured minimum level, the ERP automatically triggers:
 - Dashboard alerts
 - Low-stock warnings in inventory screens
 - Reorder recommendations
 - Inclusion in centralized procurement review reports
 - Low-stock items are visually highlighted to support quick identification and action.
- Centralized Review Authority

All items that fall below minimum order level appear in a centralized review list accessible to:

Purchase Department

CTO (Chief Technical Officer)

Authorized reviewers can:

- Evaluate urgency and consumption trends
- Verify technical relevance of replenishment
- Consolidate purchase planning across branches
- Initiate or approve procurement actions

This shared oversight ensures both operational responsiveness and executive control.

Business Benefits

- **Prevents stockouts and lost sales**
- **Enables proactive replenishment planning**
- **Reduces emergency purchases**
- **Improves centralized procurement efficiency**
- **Strengthens executive inventory governance**
- **Supports branch-level accountability**

Summary:

Minimum Order Level control transforms inventory management from reactive to proactive. By combining automated alerts with centralized review by the Purchase team and CTO, the ERP ensures continuous stock availability, disciplined procurement, and data-driven inventory planning across all branches.

8.10. Item Handling Warnings and Sales Alerts

The proposed ERP will include an Item Handling Warning and Alert mechanism to guide sales personnel during customer transactions. Certain items may require special handling instructions or precautionary actions before delivery. These operational notes will be stored in the Item Master and automatically displayed as alerts during sales processing.

Key Features

Item-Specific Warning Messages: Each item can contain predefined operational warnings such as:

- Remove sticker before delivery
- Remove original packaging/box
- Verify compatibility before sale
- Special handling instructions
- Warranty or return conditions

Automatic Sales Alerts: When the item is selected in a sales transaction, the system displays a pop-up warning or highlighted alert. The salesperson must acknowledge the alert before proceeding.

Mandatory Acknowledgement Option (Configurable if required): For sensitive items, the ERP can require confirmation to ensure instructions are followed.

8.11. External Parts Reference Integration – Part Souq

Currently, employees refer to PartSouq.com, a third-party online parts catalogue, to verify part details, OEM numbers, pricing, and compatibility during sales and procurement activities. While useful, accessing an external website creates workflow interruptions, manual effort, and dependency on external systems, which the proposed ERP aims to minimize.

Enhancement Options

- API Integration (Preferred, if available)
- Direct integration with PartSouq.com via API will allow ERP users to search and retrieve parts information within the ERP interface.
- Search results can include OEM numbers, alternative brands, technical specifications, pricing, and availability.
- Enables real-time access to updated catalogue data without leaving the ERP environment.
- Embedded Browser Access (Fallback Option)
- If API integration is not available, the ERP will provide an embedded browser window to access PartSouq.com within the system interface.

Users can browse the external catalogue without leaving the ERP, preserving workflow continuity.

Search results can be referenced manually to update ERP item details while maintaining operational control.

Objectives and Benefits

- **Maintain Workflow Continuity:** Employees remain inside the ERP while accessing external reference data.
- **Reduce Manual Effort:** Eliminates repeated switching between ERP and external sites, reducing errors and delays.
- **Improve Data Accuracy:** Direct or embedded access allows for more accurate item verification.
- **Support Substitute Recommendations:** Enables mapping and substitution logic based on verified catalogue data.
- **Centralize Knowledge:** Consolidates external reference data usage within ERP analytics and reporting.

8.12. Item Mapping – Substitute and Compatibility Intelligence

Current Item Mapping Process

1. In the existing system, item mapping is handled manually within the Item Master. Specifically:
2. OEM numbers of compatible or substitute parts are entered in the comments section of the Item Master.
3. This allows sales and inventory staff to identify potential alternatives when the primary item is

unavailable.

4. The mapping is informational only, and there is no automated link between the base item and substitutes.

• **Proposed ERP – Item Mapping Window**

The proposed ERP introduces a dedicated Item Mapping Window to replace the manual comment-based OEM mapping in the current system. This window allows users to efficiently and accurately link items with their compatible OEM numbers through a structured interface.

Key Features:

Suggested OEM Numbers:

- The system automatically suggests potential OEM numbers for mapping based on existing database entries, item specifications, and historical mappings.
- Users can select from these suggestions with a single click to map them to the item.

Manual OEM Selection:

If the suggested OEM numbers are not suitable, users can manually add OEM numbers directly within the mapping window.

This ensures all valid alternatives are captured, even for newly introduced parts.

Multiple Mappings:

A single item can be mapped to multiple OEM numbers, allowing for substitutes, brand alternatives, and cross-reference compatibility.

Integration with Item Master:

Mapped OEM numbers are structurally linked to the base item rather than stored in a free-text comments field.

This enables the ERP to automatically include these mappings in search results, substitute suggestions, and sales workflows.

Governance and Security:

- Only authorized personnel (e.g., CTO or Inventory Manager) can modify mappings.
- All mapping changes are version-controlled and audit-tracked for compliance and data integrity.

Business Benefits:

- Eliminates reliance on manual comments for mapping
- Reduces errors and omissions
- Enhances sales efficiency by providing automated substitute suggestions
- Supports accurate inventory reporting and decision-making

- Preserves internal trade intelligence and mapping confidentiality

Summary:

The ERP Mapping Window transforms item mapping from a manual, unstructured process into a controlled, automated, and auditable workflow, enabling seamless substitute recommendation, operational efficiency, and data-driven decision-making.

8.13. Enhanced Sales Support through Advanced Searching and Mapping

The ERP's Item Mapping Window, combined with the Advanced Substitute Suggestion Engine, empowers sales personnel with advanced search capabilities. By leveraging mapped OEM numbers and item attributes, sales staff can:

- Quickly identify alternative or compatible items that meet customer requirements
- Suggest substitutes across different brands or pricing tiers
- Respond promptly to customer demands, even when the requested item is out of stock

Business Impact:

- Ensures higher sales conversion by providing relevant options instantly
- Improves customer satisfaction and retention through timely and accurate recommendations
- Reduces lost sales opportunities due to stock shortages
- Strengthens the company's competitive positioning by offering flexibility and informed alternatives

Summary:

By integrating structured OEM mapping and intelligent search functionality, the ERP turns item compatibility data into actionable insights, allowing sales personnel to make informed recommendations that align with customer needs and business objectives.



8.14. Executive Item Audit Visibility

CTO-level users receive extended audit analytics:

- Branch-wise stock visibility
- Purchase history
- Sales history
- Price trends
- Movement analysis
- Demand insights

This supports executive inventory governance.

Area	Current System	Proposed System
PCode Generation	Manual in Excel, bulk upload to ERP; sequence errors; high dependency on individuals	Automated system-generated PCode; sequence verification; item-wise save; permanent PCode lock; CTO approval workflow
Price Management	Price updates done manually per country; time-consuming; risk of inconsistencies	Centralized pricing interface; bulk update across countries; currency conversion; audit logs; country-level overrides
Inventory Control	No formal minimum stock control; branch-wise visibility limited; stock alerts manual	Minimum order level per branch/store; automated low-stock alerts; centralized review by Purchase & CTO
Item Mapping / Substitutes	OEM numbers in comment field; mapping unstructured; no automated suggestions	Structured Item Mapping Window; automated substitute suggestions; brand & price-tier alternatives; mapping access restricted
Exception Handling / Emergency Sales	Ad-hoc handling by sales or store; temporary manual tracking	Dedicated non-PCode master; CTO review; sequential PCode generation; temporary transactions linked to official record
Sales Alerts / Warnings	No structured warnings; risk of operational errors	Item-specific handling alerts (e.g., remove sticker/box); mandatory acknowledgment option
External Reference Catalog	Employees use PartSouq.com manually; workflow interrupted	ERP integration via API (preferred) or embedded browser; maintains workflow inside ERP
Legacy / Historical Data	Old/unused PCodes searchable; cluttered master; reduced search performance	Structured cleanup; inactive PCodes hidden; audit-archived; performance improved
Executive Visibility / Analytics	Limited CTO visibility; reporting manually compiled; branch data fragmented	Extended CTO audit dashboard: stock, sales, purchase, pricing trends, movement analysis
Overall Governance	High dependence on individuals; inconsistent process; audit and control gaps	Centralized approval workflows, role-based access, audit tracking, exception control, master data governance

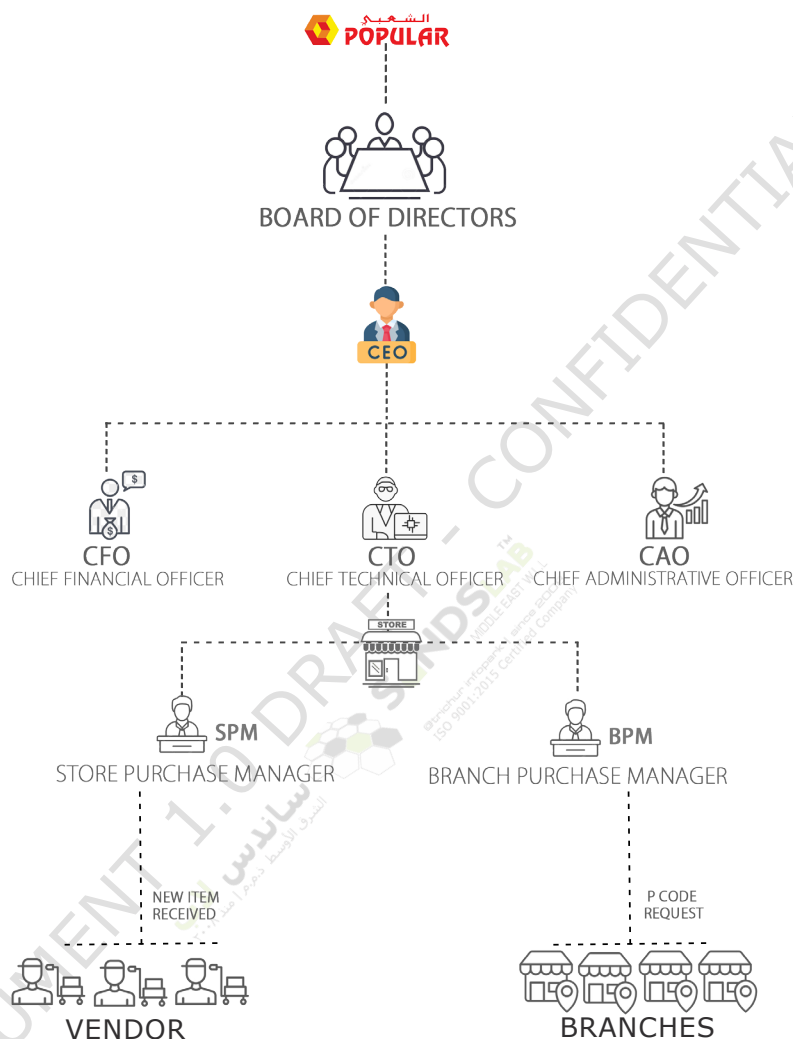


9. Management Review Points – Current vs Proposed System

Summary of Key Management Review Decisions

1. Approve automated PCode generation sequence and exception handling rules.
2. Confirm CTO and Purchase Department authority scope for approvals and exceptions.
3. Define minimum stock levels per branch/store and review mechanism.
4. Approve centralized price management process and audit requirements.
5. Confirm Item Mapping & substitute recommendation strategy and access restrictions.
6. Decide on integration approach for PartSouq catalog (API vs embedded link).
7. Approve legacy PCode cleanup policy and archival process.
8. Define standard sales warnings and mandatory acknowledgment process.
9. Approve executive dashboard metrics, reporting frequency, and visibility rules.
10. Confirm overall governance, audit, and exception control policies for inventory operations.

PCODE GENERATION FLOW



10. PCode Generation Process

The system already contains a large number of PCodes, some of which are actively used and others that may be inactive. Therefore, a controlled process is required for generating new PCodes to avoid duplication and maintain data accuracy.

Case 1: Request from Sales Counters

At different branch sales counters, users may identify that a required item is not available in the main store. In such cases, the user searches using the OEM number or part number.

If the system does not contain that specific OEM or part number, the user submits a request for a new PCode to the Branch Purchase Manager (BPM).

The BPM will then verify the entire system to check whether the same OEM or part number already exists under a different brand, company, or description.

If the item already exists, the BPM will map the request to the existing PCode.

If the BPM confirms that it is a completely new item, the request will be forwarded to the

Chief Technical Officer (CTO).

- The CTO will verify the OEM or part number using:
- The existing system records
- Domain knowledge
- External reliable sources

If the CTO confirms that it is a new item, a new PCode will be generated for that specific OEM or part number.

All these steps will be tracked within the system. Once the item is registered in the main store database, the new OEM or part number will automatically become available to all branches and sales counters, since the system operates on a cloud infrastructure.

Case 2: Items Received from Vendors

In another scenario, the CTO may place orders with vendors. When the items are received, the Store Purchase Manager (SPM) will verify the received items against the vendor invoice and the original purchase order created by the CTO.

If some items are supplied with a different brand than expected, the SPM will send a request to the CTO to generate a new PCode for those items.

The CTO will then:

- Verify the item details
- Confirm whether it is a new item
- Generate a new PCode if required

Once the new PCode is generated, it will automatically reflect across all branches and stores in the system.

This process ensures:

- **Centralized control over item codes**
- **Prevention of duplicate PCodes**
- **Standardized item identification across all branches**
- **Real-time availability of new items throughout the organization**

End of Session